VLM Reliability Toolkit: Unified Inference-Time Hallucination Detection and Attribute-Binding Correction for Vision-Language Models

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***Abstract*—Vision-language models (VLMs) generate fluent image descriptions but can mention objects that are absent from the image and bind attributes to the wrong objects. These failures, known as object hallucination and attribute-binding errors, call for complementary verification, because confirming that an object is present does not establish that its attributes are correctly assigned. This paper presents the VLM Reliability Toolkit, a training-free, model-agnostic framework that detects and corrects both failure types in a single inference-time pipeline. The framework decomposes generated text into atomic claims with a deterministic rule-based extractor, verifies object existence with open-vocabulary grounding, and verifies attributes on the detected region of the object itself. A reliability score that combines visual evidence, token confidence and inter-candidate agreement then drives a per-claim accept, correct or abstain policy and an answer-level re-ranking rule, with corrections applied only when positive visual evidence supports the replacement. On LLaVA-1.5-7B, the toolkit lowers $CHAIR_S$ from 63.8% to 57.2% and $CHAIR_I$ from 29.6% to 25.5% on 500 COCO images, while object recall changes by 0.8 points. On POPE, accuracy stays within 0.1 points of the baseline while precision rises by up to 0.9 points. The full pipeline costs 2.67× the latency of greedy decoding. Ablations show that single-candidate claim correction yields the larger share of the CHAIR gain and that reliability-guided re-ranking contributes the remainder. All results can be reproduced on a CPU from cached verifier outputs.**

Keywords - Vision-language models, object hallucination, attribute binding, claim verification, post-hoc correction, reliability scoring

I. Introduction

Vision-language models (VLMs) such as LLaVA [1], [2], BLIP-2 [3], InstructBLIP [4] and MiniGPT-4 [5] have advanced image understanding and natural-language generation, supporting applications in visual search, accessibility and image-based question answering. However, their outputs may contain statements that are not supported by the image, which reduces reliability wherever accurate descriptions matter [6].

Two failure modes are of particular interest. Object hallucination occurs when a model mentions an object that is absent from the image [6]–[8]. Attribute-binding errors occur when an attribute is assigned to the wrong object even though the relevant objects and attributes are present: a model may see a red cup on a blue table but describe a blue cup on a red table. CHAIR and POPE measure object hallucination [7], [8], whereas ARO, Winoground and SugarCrepe probe compositional understanding and object–attribute association [9]–[11]. The two failure modes call for complementary verification, because confirming that an object is present does not establish that its attributes are correctly assigned.

Existing mitigation methods include trained revision, decoding-time strategies and post-hoc correction. LURE trains a revisor to rewrite hallucinated descriptions [12], VOLCANO uses self-feedback-guided revision [13], OPERA penalises over-trust during decoding [14], and Woodpecker corrects responses with external visual perception models [15]. These methods form a strong foundation. A remaining gap is a transparent claim-level policy that verifies both object existence and attribute binding, edits only what the evidence contradicts, and abstains when the evidence is insufficient. This motivates the question: **how can both error types be detected and selectively corrected within one training-free, inference-time framework while preserving supported content?**

We propose the VLM Reliability Toolkit, a training-free, model-agnostic reliability layer that runs after any VLM. It decomposes each answer into atomic claims, verifies object existence with open-vocabulary grounding, verifies attributes on the object's own detected region, and then scores each candidate answer and applies a per-claim accept, correct or abstain policy together with an answer-level re-rank rule. No language model judges another language model, and verified

text is never rewritten. On LLaVA-1.5-7B the toolkit reduces $CHAIR_S$ from 63.8% to 57.2% and $CHAIR_I$ from 29.6% to 25.5%, while object recall changes by less than one point. The main contributions are as follows:

- Claim-level verification. A pipeline that separates object hallucination from attribute-binding errors and binds each attribute to its object's detected region (Section IV-D).
- Reliability scoring. A score that combines visual evidence, token confidence and inter-candidate agreement, with unavailable terms redistributed rather than invented (Section IV-E).
- Evidence-gated decisions. A policy that corrects only with positive visual evidence, abstains otherwise, and re-ranks across n candidates (Section IV-F).
- Reproducible evaluation. A two-stage protocol (GPU generation, then CPU reprocessing from cached verifier outputs) on POPE and CHAIR with five ablations, together with a discussion of where the method does not help (Sections V and VI).

II. Related Work

Benchmarks. Rohrbach et al. introduced CHAIR, which measures object hallucination in captions by comparing mentioned objects with ground-truth annotations [7]. Li et al. proposed POPE, which recasts object-presence evaluation as a set of yes/no polling questions [8]. HallusionBench diagnoses language-related bias and visual misinterpretation in large VLMs [16]. For compositionality, ARO examines attribute, relation and order understanding [9], Winoground requires a model to match two captions with two images [10], and SugarCrepe uses hard negative captions designed to remove exploitable artefacts of earlier benchmarks [11].

Mitigation. LURE trains a revisor that rewrites hallucinated descriptions [12]. VOLCANO has the model critique and revise its own answer through self-feedback [13]. OPERA mitigates hallucination during decoding by penalising over-trust and allowing retrospection [14]. Woodpecker is a training-free, post-hoc framework that validates concepts with external visual models and corrects the response accordingly, and relies on language models for some of its stages [15]. Our toolkit is likewise training-free and post-hoc, but keeps extraction, verification, scoring and editing deterministic and rule-based, represents attribute binding explicitly, and can abstain.

III. Problem Formulation

Let I be an image, q a text query and $y = M(I, q)$ the response of a VLM M. We parse y into a set of atomic claims C(y). A claim is an existence claim $e = (o)$, asserting that object o is present; an attribute claim $a = (o, k, v)$, asserting that o has value v for attribute category k (colour or material); a quantity claim (o, m); or a relation claim (o_1, r, o_2). Let $z(c) \in \{0, 1\}$ denote whether claim c is true of I according to ground-truth annotation.

An existence claim with $z = 0$ is a hallucination. An attribute claim $a = (o, k, v)$ is a binding error when the object is present, the claim is false, and the asserted value is true of a different object o′ in the same image:

$$o \in I, \quad z(o, k, v) = 0, \quad \exists\ o' \neq o : z(o', k, v) = 1 \quad (1)$$

An attribute claim that is false without satisfying (1) is a plain attribute error, since no object in the scene carries the asserted value. The distinction matters because the cases call for different repairs: a hallucinated object should be removed, a swapped attribute should be re-bound, and a plain attribute error should be replaced by the observed value. The goal is a response y′ that removes false claims while preserving the true claims in y, so correction must not be achieved by deleting content indiscriminately.

IV. Proposed Method

A. System Architecture

The toolkit (Fig. 1) comprises seven cooperating components: a VLM execution engine that loads the base model; an inference orchestrator that requests n candidate answers; a diagnostic engine that extracts and verifies claims; a reliability and scoring engine; a correction and decision layer; an evaluation and benchmarking module; and a dashboard and reporting module for human review. The system runs entirely at inference time and treats the VLM as a black box, or as a white box when token probabilities are available.

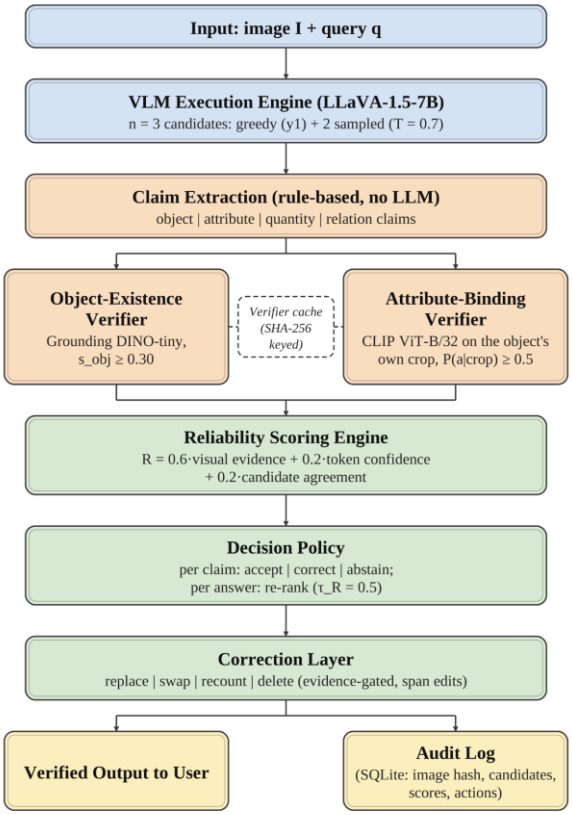


Fig. 1. Inference-time diagnostic and correction pipeline of the VLM Reliability Toolkit.

B. Candidate Generation

For each image and query the VLM produces n candidates (n = 3 for captioning). Candidate 1 uses greedy decoding and is the uncorrected baseline; candidates 2 and 3 are sampled at temperature 0.7 with seed 42 and at most 128 new tokens. Local (white-box) models, namely LLaVA-1.5-7B, LLaVA-OneVision-0.5B, LLaVA-Next and BLIP-2, draw the sampled candidates from one generate call with n−1 return sequences. API (black-box) models (Gemini, OpenAI, Groq and OpenAI-compatible endpoints) receive independent calls issued in parallel, with temperature 0 for candidate 1. A failed generation keeps its slot with the diagnosis GENERATION_FAILED, so candidate 1 can only be replaced through the re-rank rule.

For white-box models the token confidence of candidate j is the length-normalised sequence probability, computed from the raw logits of the generated tokens with padding removed:

$$p_j = \exp[\ (1/T_j)\ \Sigma_t \log P(w_t \mid w_{<t}, I, q)\ ] \quad (2)$$

where the sum runs over the T_j generated tokens. API models expose no token probabilities (OpenAI log-probabilities are used when present), so p_j is marked unavailable rather than estimated.

C. Claim Extraction

Each candidate is split into sentences and parsed by a deterministic, rule-based extractor; no language model is used. Every claim keeps its character span in the original text, which the correction step later edits. Table I lists the claim types. Objects are the 80 COCO classes [17] with the CHAIR synonym list (e.g., "man" and "woman" map to person) [7], plus common clothing and scene nouns. Attributes are 11 colours (red, blue, green, yellow, black, white, orange, brown, gray, purple, pink) with aliases and 8 materials (wooden, metal, plastic, glass, leather, stone, fabric, paper); an attribute binds to the noun it modifies. For relations, the nearest following object after the relation phrase is the target. The extracted claims also form a claim graph (objects, instances, attribute values) that is shown in the dashboard.

Table I .Claim Types Handled by the Extract

Claim type	Form	Example
Object (existence)	⟨o⟩	"a dog" → dog exists
Attribute (binding)	⟨o, a⟩, a ∈ colour ∪ material	"a red shirt" → shirt is red
Quantity	⟨o, m⟩	"two dogs" → count(dog) = 2
Relation	⟨o_1, r, o_2⟩, r ∈ {left of, right of, above, below, next to, near}	"a cat near a car"

D. Visual Verification

Every claim receives a verification score $v_i \in [0, 1]$ and a status (SUPPORTED, CONTRADICTED, UNSUPPORTED or AMBIGUOUS), computed only from detector boxes and region-level CLIP probabilities.

- Existence. Grounding DINO-tiny [18] is queried with the object noun as an open-vocabulary prompt. Let B(o) be the set of returned boxes whose detection confidence d_b is at or above a floor of 0.10, so that weak evidence remains a real score instead of being clipped to zero:

$$s_{obj}(o) = \max_{b \in B(o)} d_b \quad (3)$$

- The claim is SUPPORTED when $s_{obj} \geq 0.30$ (the grounding threshold) and otherwise UNSUPPORTED, with diagnosis OBJECT_HALLUCINATION. A low detection score is not by itself proof that an object is absent, particularly for small or occluded objects (Section VII).
- Attribute binding. For each of the (at most five) most confident boxes of the object, the box is cropped and CLIP ViT-B/32 [19] scores a closed label set with prompts of the form "a photo of a {colour} {object}" (for materials, "a photo of a {object} made of {material}"). The binding score multiplies the detection confidence and the crop probability on the same region:

$$s_{attr}(o, a) = \max_{b \in B(o)} d_b \cdot P(a \mid crop_b) \quad (4)$$

- The claim is SUPPORTED if $P(a \mid crop) \geq 0.5$ on the best region. It is CONTRADICTED if the most confident instance's top label $a' \neq a$ has $P(a' \mid crop) \geq 0.5$, in which case a' is recorded as the verified value (diagnosis ATTRIBUTE_BINDING_ERROR). Otherwise it is UNSUPPORTED, and always so when the object was not localised ($v_i = 0$). Because only the object's own crop is probed, colour elsewhere in the image cannot support the claim. CLIP probabilities express relative support among the configured labels and are not calibrated probabilities of truth.
- Quantity and relation. For quantity claims, boxes at or above the grounding threshold are de-duplicated by greedy non-maximum suppression (IoU 0.5); $v_i = s_{obj}$ if the surviving count equals the claimed m and 0 otherwise, and the observed count is recorded. For relation claims, box-centre geometry decides left of, right of, above and below; next to and near hold when the L1 centre distance is below twice the larger box's width plus height. Then $v_i = \min(s_{obj}(o_1), s_{obj}(o_2))$ if the geometry holds and 0 otherwise, and the claim is AMBIGUOUS if either object is not localised. AMBIGUOUS means that the verifier could not run (a model was unavailable or the attribute type is unsupported); such claims are excluded from scoring and never auto-corrected.

E. Reliability Scoring

Each candidate j receives one score $R_j \in [0, 1]$. With fixed weights $\lambda_1 = 0.6$, $\lambda_2 = 0.2$ and $\lambda_3 = 0.2$, the score is computed over the set $U \subseteq \{1, 2, 3\}$ of terms that are available for that candidate:

$$R_j = (\ \Sigma_{t \in U}\ \lambda_t\ x_{j,t}\ )\ /\ (\ \Sigma_{t \in U}\ \lambda_t\ ) \quad (5)$$

Here $x_{j,1}$ is the mean verification score over the claims of candidate j whose verifier ran (AMBIGUOUS claims excluded), $x_{j,2} = p_j$ is the token confidence of Section IV-B, and $x_{j,3} = A_j$ is the fraction of its claims also asserted by at least one of the other n−1 candidates, where claims match exactly on type, object, attribute, value and relation target. A term is unavailable when no claim could be verified ($x_{j,1}$), for API models (p_j) and when n = 1 (A_j). An unavailable term is never imputed; its weight is redistributed proportionally over the remaining terms, and when all three terms are available the denominator is 1. For example, a typed caption with four claims scoring 0.79, 0.88, 0.81 and 0.81 (mean 0.8225), $A_j = 1.0$ and no token probability obtains $R_j = (0.6 \times 0.8225 + 0.2 \times 1.0) / 0.8 = 0.867$. The score R_j is used only at the answer level, for re-ranking and whole-answer abstention; per-claim actions use each claim's own v_i.

F. Decision Policy, Re-ranking and Correction

The policy acts per claim and then per answer. A correction always requires positive evidence for the replacement; without it the claim is flagged and left unchanged. Table II gives the per-claim rule with type-specific thresholds $\tau_{lo} < \tau_{hi}$ (Section V-B).

TABLE II : Per-Claim Decision Policy

Condition on v_i	Action	Edit
Verifier did not run (AMBIGUOUS)	Abstain	none
$v_i \geq \tau_{hi}$	Accept	none
$v_i \leq \tau_{lo}$, object, $s_{obj} <$ 0.30	Correct	delete the object mention
$v_i \leq \tau_{lo}$, attribute or quantity, verified value exists	Correct	replace with the observed value, or swap
$v_i \leq \tau_{lo}$, no substitute (e.g., relation)	Abstain	none
$\tau_{lo} < v_i < \tau_{hi}$	Abstain	none (flagged as uncertain)

At the answer level, with $\tau_R = 0.5$:

- If the greedy candidate has $R_1 < \tau_R$ and the best candidate $j^* = \text{argmax}_j R_j$ satisfies $R_{j^*} \geq \tau_R$, select j^* (RERANK).
- Otherwise keep candidate 1. If $R_1 < \tau_R$ and no CORRECT action exists, the answer is ABSTAIN (no reliable candidate); a low R alone does not trigger abstention when evidence-backed corrections were found.
- Otherwise the decision is CORRECT if any claim was corrected, ABSTAIN if any claim was flagged, and ACCEPT if neither.

Correction edits character spans of the selected candidate's original text, so verified words are never rewritten. Four operations are implemented: replace, where an attribute word becomes the verified value ("blue shirt" → "red shirt"); swap, where two attributes in a sentence that are each contradicted by the other's claimed value are exchanged ("blue shirt and red jeans" → "red shirt and blue jeans"); recount, where a number word becomes the detected count; and delete, where a hallucinated object's phrase is removed, including coordinated subjects ("a dog and a cat" → "a dog") and trailing clauses ("holding a surfboard").

V. Implementation and Experimental Setup

A. Models, Data and Metrics

All reported numbers use LLaVA-1.5-7B [2] (fp16) as the generator, Grounding DINO-tiny and CLIP ViT-B/32 as verifiers, and seed 42, on Kaggle with two NVIDIA T4 GPUs (POPE on one, CHAIR on the other).

- POPE. We use the official POPE questions on COCO val2014 images: 450 images × 3 splits (random, popular, adversarial), giving 2,700 questions per split. A further 50 images form the tuning split (Section V-B). The query is "{question} Answer the question using a single word or phrase." with 8 new tokens and one greedy candidate. An answer is "no" if "no" or "not" appears in the first sentence and "yes" otherwise; a toolkit-corrected existence claim turns "yes" into "no". We report accuracy, precision, recall, F1, yes-ratio and abstention rate.
- CHAIR. We use 500 COCO val2014 images disjoint from the tuning images, with ground truth taken from the COCO instance annotations mapped through the CHAIR synonym list. The query is "Describe this image in detail." (128 new tokens) with 3 candidates. $CHAIR_S$ is the fraction of captions with at least one hallucinated object, and $CHAIR_I$ is the number of hallucinated object mentions divided by all object mentions [7]. Object recall is the number of ground-truth objects mentioned divided by all ground-truth objects; it shows whether deletion removes real objects. We also report the mean caption length in words.

Latency and calibration. The latency ratio is the mean time to generate all candidates and verify them divided by the mean time of greedy-only generation. Calibration is measured by the expected calibration error (ECE) [20] and the Brier score of R against the event that the final answer is correct, over asserted POPE claims.

B. Threshold Tuning and Weight Selection

Thresholds are tuned once on a held-out split of 50 POPE images (300 questions, random split) that is disjoint from every reported image. On POPE a "yes" answer asserts an existence claim, and the toolkit turns it into "no" when $v_i \leq \tau_{lo}$ and $s_{obj} <$ 0.30. The threshold τ_{lo} is chosen on the grid 0.10, 0.15, …, 0.45 to maximise the F1 of the final answers (ties broken by lower abstention); τ_{hi} is the smallest value above τ_{lo} (step 0.05) at which accepted claims ($v_i \geq \tau_{hi}$) reach 95% precision, with at least 10 accepted claims. The result is $\tau_{lo} = 0.15$ and $\tau_{hi} = 0.20$ (F1 = 0.848 and abstention 0.67% on the tuning split), applied to object, quantity and relation claims. Attribute thresholds keep their configured values ($\tau_{lo} = 0.15$, $\tau_{hi} = 0.40$), because POPE contains no attribute claims.

A grid search over the share of λ_1 versus λ_2 (step 0.1) that minimises ECE on the 121 asserted claims selects $\lambda_1 = 0$ (ECE = 0.050). We do not apply it, for two reasons. First, POPE answers are a single yes/no token from one candidate, so p_j is trivially well calibrated there. Second, with $\lambda_1 = 0$ the score ignores visual evidence on captions: on CHAIR the full toolkit then reaches only $CHAIR_S = 59.4$ and $CHAIR_I = 27.0$, against 57.2 and 25.5 with the visual-dominant weights. The weights therefore remain at $\lambda = (0.6, 0.2, 0.2)$. The unconstrained result is recorded with the tuned thresholds, and tuned weights are opt-in with λ_1 constrained to at least 0.5.

C. Ablation Variants

Table III lists the five variants. The baseline is candidate 1 of the same run, so the baseline and the toolkit share identical generations. POPE answers carry no attributes and use one candidate, so only the base VLM and the full toolkit are reported for POPE.

Table III : Ablation Variants

Variant	Verification	Attr. probe	n	Reliability / re-rank
Base VLM	off	off	1	off
Full toolkit	on	on	3	on
No attribute binding	on	off	3	on
No reliability scoring	on	on	3	off
n = 1	on	on	1	on

D. Implementation and Reproducibility

The toolkit is written in Python 3.12/3.13 with PyTorch and Hugging Face Transformers (below version 5); the service is built with FastAPI and the dashboard with Next.js. A suite of 78 unit, pipeline, API and evaluation tests runs without a GPU.

Two-stage evaluation. Generation (GPU) writes resumable JSONL files; processing reads them, runs verification and every ablation, and writes per-variant JSONL. Each processed row records its configuration (thresholds, weights, tuned flag), and the report warns when rows with different settings are mixed.

Verifier cache. Every Grounding DINO and CLIP output is cached on disk, keyed by the image's SHA-256 hash and the query (object name, box, attribute type), together with its compute time. Thresholds, weights and ablations can therefore be re-evaluated on a laptop in cache-only mode: reprocessing the full POPE (16,200 rows) and CHAIR (2,500 rows) took about 20 s each with no model loaded. Verification cost is the cached per-entry compute time, so cached and freshly computed rows report the same latency and the latency ratio does not depend on cache state.

Result sharing and audit log. Partial results are pushed to a Hugging Face dataset every 50–300 samples, so a stopped Kaggle session resumes from the last upload. Every live analysis is stored in SQLite with the image hash, all candidates, claim scores, evidence, actions and the decision, which enables regression testing and provenance review.

CPU deployment. For the dashboard on machines without a GPU, Grounding DINO runs at 480 × 800 input (about 3× faster, with scores within ±0.03 on our test image) and the local VLM sees one 384 px tile. These runs are cached separately and are never mixed into the benchmark results, which use full resolution on GPU.

VI. Results and Discussion

A. POPE: Object Existence

Table IV reports the POPE results. The toolkit changes POPE answers very little. Accuracy moves by +0.1, 0.0 and +0.1 points on the random, popular and adversarial splits, and F1 by +0.1, −0.1 and 0.0. Precision rises on every split (by 0.9, 0.4 and 0.7 points), while recall falls from 77.6% to 77.2% and the yes-ratio falls by 0.4–0.6 points, roughly 11–16 of the 2,700 answers per split. Abstention stays at 0.4–0.5%.

Table IV : POPE Results (N = 2,700 per split)

Method	Split	Acc.	Prec.	Rec.	F1	Yes %	Abst. %
Base	Random	87.6	96.9	77.6	86.2	40.0	0.0
Toolkit	Random	87.7	97.8	77.2	86.3	39.4	0.4
Base	Popular	86.6	94.7	77.6	85.3	41.0	0.0
Toolkit	Popular	86.6	95.1	77.2	85.2	40.6	0.5
Base	Adv.	84.3	89.6	77.6	83.2	43.3	0.0
Toolkit	Adv.	84.4	90.3	77.2	83.2	42.7	0.5

(Base = base VLM (LLaVA-1.5-7B); Toolkit = full toolkit; Adv. = adversarial. All values in %.)

The mechanism explains the small effect. The toolkit can only turn a "yes" into a "no", and the baseline already answers "yes" for only 40.0–43.3% of the questions of a benchmark that is designed to be balanced. Its dominant error is therefore a missed object (recall 77.6%), which verification of asserted claims cannot repair. The visual evidence removes a few false positives at a small cost in true positives.

B. CHAIR: Caption Hallucination

Table V reports the CHAIR results and Fig. 2 plots $CHAIR_S$ against latency. The full toolkit lowers $CHAIR_S$ from 63.8% to 57.2% (−6.6 points, 10.3% relative) and $CHAIR_I$ from 29.6% to 25.5% (−4.1 points, 13.9% relative). Object recall falls by 0.8 points (76.2% to 75.4%) and the mean caption length by 2.5 words (86.4 to 83.9), so the reduction is not obtained by removing large amounts of content. The price is latency: the full pipeline takes 2.67× as long as greedy decoding.

Table V : CHAIR Results and Ablations (N = 500 Images)

Variant	$CHAIR_S$	$CHAIR_I$	Recall	Words	Latency
Base VLM	63.8	29.6	76.2	86.4	1×
Full toolkit	**57.2**	**25.5**	**75.4**	**83.9**	**2.67×**
n = 1	59.6	27.2	75.8	84.4	1.23×
No reliability scoring	59.6	27.2	75.8	84.4	2.67×
No attribute binding	57.6	26.0	75.5	83.9	2.67×

$CHAIR_S$, $CHAIR_I$ and recall in % (lower CHAIR is better); Words = mean caption length; Latency relative to greedy decoding.

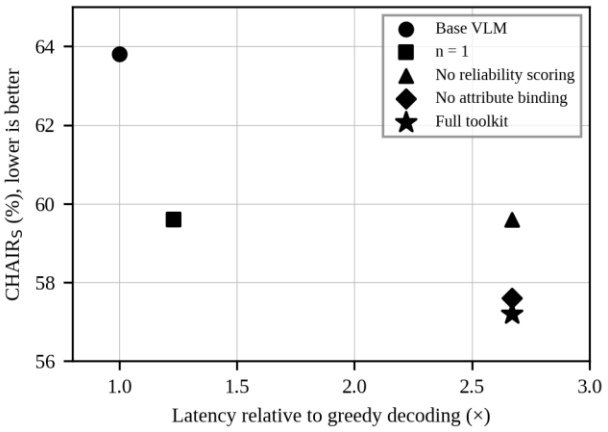


Fig. 2. $CHAIR_S$ versus latency for the five variants of Table V (lower left is better).

C. Ablation Analysis

Single-candidate correction (n = 1) already lowers $CHAIR_S$ by 4.2 points and $CHAIR_I$ by 2.4 points at only 1.23× latency, so the larger share of the gain comes from claim-level correction itself. Adding two sampled candidates with reliability-guided re-ranking lowers both metrics by a further 2.4 and 1.7 points, at an additional 1.44× of latency.

Disabling reliability scoring reproduces the n = 1 numbers exactly (59.6 and 27.2). Without a score, candidate 1 is always kept, so the additional candidates are generated but never used, yet they still incur the full 2.67× latency. Re-ranking is therefore the only route through which extra candidates help in this setting.

Removing the attribute probe changes the results only slightly (57.6 and 26.0 versus 57.2 and 25.5; for $CHAIR_S$ this is two captions out of 500). This is expected, because CHAIR ground truth contains objects but no attributes, so the benchmark cannot measure the contribution of binding verification. We regard that contribution as untested in this paper (Section VII).

D. Illustrative Example

To illustrate the correction behaviour, consider an image of a person on a beach wearing a red shirt and blue jeans, with the query "Describe this image in detail." The VLM answers: "A person in a blue shirt and red jeans stands on the beach holding a surfboard." Table VI traces the claims through verification. The two attribute claims are each contradicted by the region of the other, so the policy swaps the bound colours; the surfboard has no detection, so its clause is deleted. The corrected output is "A person in a red shirt and blue jeans stands on the beach." Verified content is left intact. This example is constructed to demonstrate the policy and is not a measured result.

Table VI : Claim Trace for the Illustrative Example

Claim	Verification result
person exists	pass
shirt is blue	fail: region is red
jeans are red	fail: region is blue
surfboard exists	fail: not detected
beach exists	pass

VII. Limitations

- Correction quality is bounded by the verifier: small or occluded objects may be missed, which leaves hallucinations uncorrected or deletes real objects, and rare categories may fall outside the detector's vocabulary.
- Claim-extraction errors propagate downstream, and the closed vocabulary covers only 11 colours and 8 materials.
- Colour is sensitive to lighting and ambiguous boundaries.
- Attribute-binding correction is not quantitatively evaluated here: POPE and CHAIR contain no attribute ground truth, so Winoground, SugarCrepe and a human-audited attribute-binding error rate remain to be run.
- On POPE the gain is negligible, because the toolkit can only convert "yes" to "no".
- All results come from one generator (LLaVA-1.5-7B), one seed and one run, without confidence intervals or significance tests; the 6.6-point $CHAIR_S$ difference on 500 captions should be confirmed with paired resampling and other generators. (vii) Object thresholds are tuned on POPE and reused for captions.
- The reliability score is not a calibrated confidence beyond the ECE analysis of Section V-B and needs human-annotated validation before it can serve as a production confidence measure.
- Post-generation correction adds latency (2.67× in the full configuration). Reasoning errors, toxicity and adversarial robustness are outside the present scope.

VIII. Conclusion

We presented the VLM Reliability Toolkit, a training-free, model-agnostic reliability layer that verifies claims extracted from VLM answers, binds attributes to detected object regions, and applies an evidence-gated accept, correct or abstain policy with candidate re-ranking. On LLaVA-1.5-7B it reduces $CHAIR_S$ by 6.6 points and $CHAIR_I$ by 4.1 points with less than one point of object-recall loss, at 2.67× the latency of greedy decoding, whereas on POPE it leaves accuracy essentially unchanged. Ablations show that single-candidate correction provides the larger share of the improvement and that reliability-guided re-ranking provides the rest. Future work will evaluate attribute binding directly on Winoground, SugarCrepe and a human-audited binding-error rate, add significance testing and further generators, reduce latency, and extend verification to richer relational and counting claims.

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